

VHC COMMUNICATION SCENARIO SPECIFICATION

Version	Document			Model
1.0.0	VHC COMMUNICATION SCENARIO SPECIFICATION			
Date	Drawn	Checked	Approved	Page

REVISION HISTORY

VER	DATE	DESCRIPTION	Note
2.1	2018-03-02	Cassette Type added to S2F49. Port Type added to Current Port States of S1F4. Scan Command Scenario added.	
2.3	2018-03-07	Add abnormal SCAN scenario. VehicleCarrierInstalled/Removed added.	
2.5	2018-03-10	SCAN scenario changed. Buffer transfer scenario added.	
2.6	2018-03-28	Transfer command comment added	
2.9	2018-08-22	Double Storage / Empty Retrieval added	
3.1	2018-09-10	Wait In / Wait Out Event Added	
3.2	2018-11-05	Fix Abort(After CarrierInstalled)	
3.3	2019-01-02	Abort Command(Normal / Global Duplicate)	
3.4	2020-08-18	Reading Station Scenario	
3.5	2022-06-08	Vehicle Empty Retrieval changed.	

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1. CONTROL STATUS

1.1. OFFLINE → ONLINE (By Host, TSC State = Paused)

Host		TSC	Comments
	Select.req	Select.rsp	* Online Attempt From MCS
Establish Communications Request	S1F13→	Establish Communications ACK	<A[3] MDLN 'MCS'>
		←S1F14	<A[3] MDLN 'RCP'>
Request ON-LINE	S1F17→	ON-LINE Acknowledge	* Mode Change by Host
		←S1F18	<ONLACK=0>
		←S6F11	* ControlState [Remote]
ACK	S6F12→	OnlineRemote (CEID:3, REPORTID:1)	
		←S6F11	* TSCState [Init]
ACK	S6F12→	TSCAutoInitiated (CEID:54, REPORTID:1)	
		←S6F11	* TSCState [Paused]
ACK	S6F12→	TSCPaused (CEID:56, REPORTID:1)	(Vehicle is not Active, Command is stored in Queue)
Date and Time Set Data	S2F31→	Date and Time Set ACK	* TSC sets date and time
		←S2F32	
Selected Equipment Status Request	S1F3→	Selected Equipment Status Data	* VID : 4, 53, 118, 91, 6, 73, 76, 254, 371, 7
		←S1F4	
Selected Equipment Status Request	S1F3→	Selected Equipment Status Data	* VID : 73
		←S1F4	> 1: TSC Init, 2: Paused, 3: Auto, 4: Pausing <TSCtate=2>

Host		TSC	Comments
Host Command (RESUME)	S2F41→		* Paused → Auto change Send
	←S2F42	ACK (HCACK=0x04 / 0x07)	* 4 = Confirmed, 7 = Already Auto
	←S6F11	TSCAutoCompleted (CEID:53, REPORTID:1)	* TSCState [Auto]
ACK	S6F12→		※ reference VID 4 : Alarm Set VID 53 : ActiveVehicles VID 118 : CurrentPortState VID 91 : EnhancedCarriers VID 6 : ControlState VID 73 : TSCState VID 76 : EnhancedTransfers VID 254 : UnitAlarmStatList VID 371 : CstInfoList VID 7 : EventsEnabled

1.2. OFFLINE → ONLINE (By Host, TSC State = Auto)

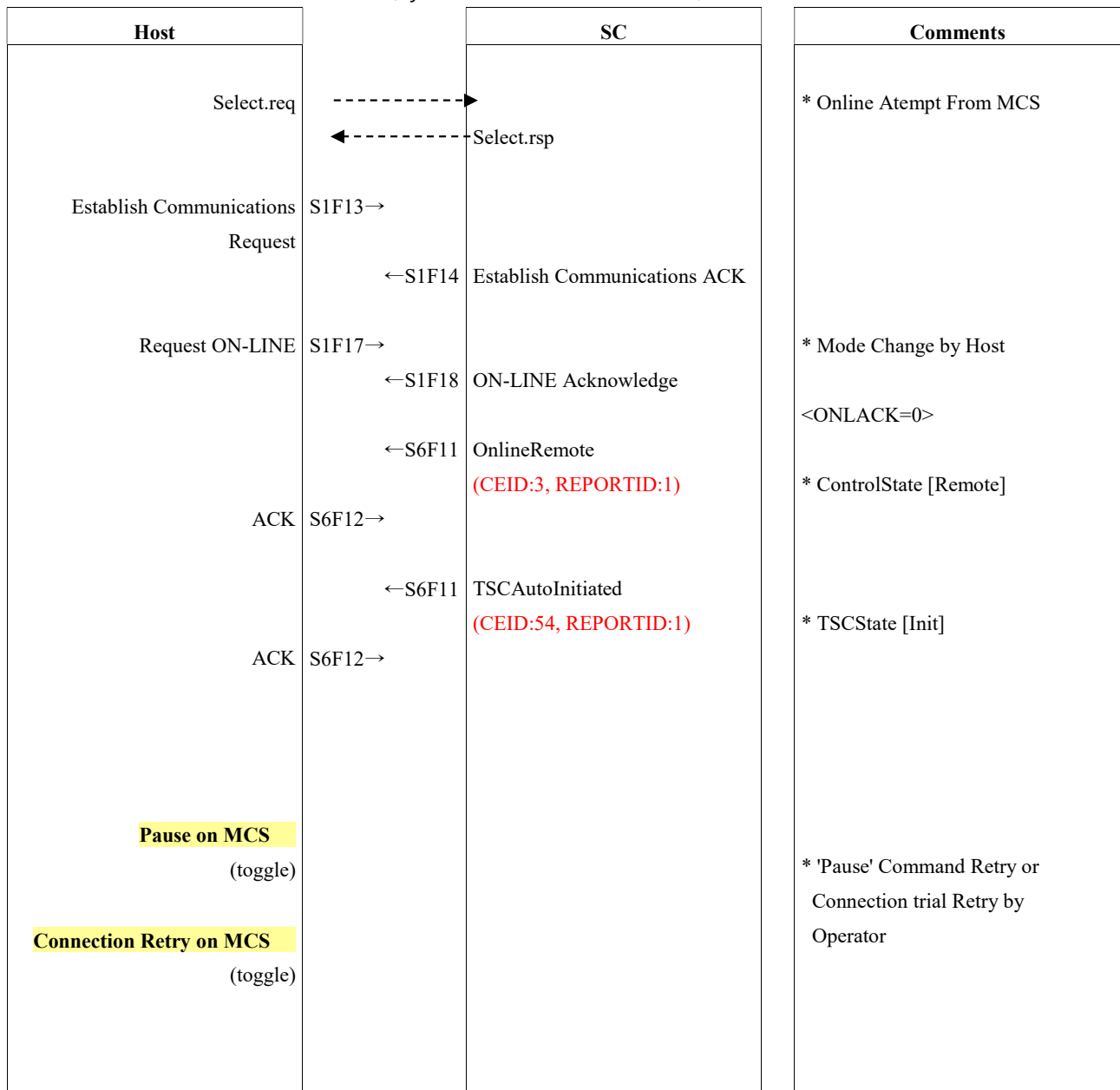
Host		TSC	Comments
	Select.req	Select.rsp	* Online Attempt From MCS
Establish Communications Request	S1F13→	←S1F14 Establish Communications ACK	<A[3] MDLN 'MCS'> <A[3] MDLN 'RCP'>
Request ON-LINE	S1F17→	←S1F18 ON-LINE Acknowledge	* Mode Change by Host <ONLACK=2>
Host Command (PAUSE)	S2F41→	←S2F42 ACK (HACK=0x04)	
		←S6F11 TSCPauseInitiated (CEID:57, REPORTID:1)	* TSCState [Pausing]
ACK	S6F12→	←S6F11 TSCPauseCompleted (CEID:55, REPORTID:1)	* TSCState [Paused] (Vehicle is not Active, Command is stored in Queue)
Date and Time Set Data	S2F31→	←S2F32 Date and Time Set ACK	* TSC sets date and time
Selected Equipment Status Request	S1F3→	←S1F4 Selected Equipment Status Data	* VID : 4, 53, 118, 91, 6, 73, 76, 254, 371, 7
Selected Equipment Status Request	S1F3→	←S1F4 Selected Equipment Status Data	* VID : 73
			> 1: TSC Init, 2: Paused, 3: Auto, 4: Pausing <SCState=2>

Host		TSC	Comments
Host Command (RESUME)	S2F41→		* Paused → Auto change Send
	←S2F42	ACK (HCACK=0x04 / 0x07)	* 4 = Confirmed, 7 = Already Auto
	←S6F11	TSCAutoCompleted (CEID:53, REPORTID:1)	* TSCState [Auto]
ACK	S6F12→		※ reference VID 4 : Alarm Set VID 53 : ActiveVehicles VID 118 : CurrentPortState VID 91 : EnhancedCarriers VID 6 : ControlState VID 73 : TSCState VID 76 : EnhancedTransfers VID 254 : UnitAlarmStatList VID 371 : CstInfoList VID 7 : EventsEnabled

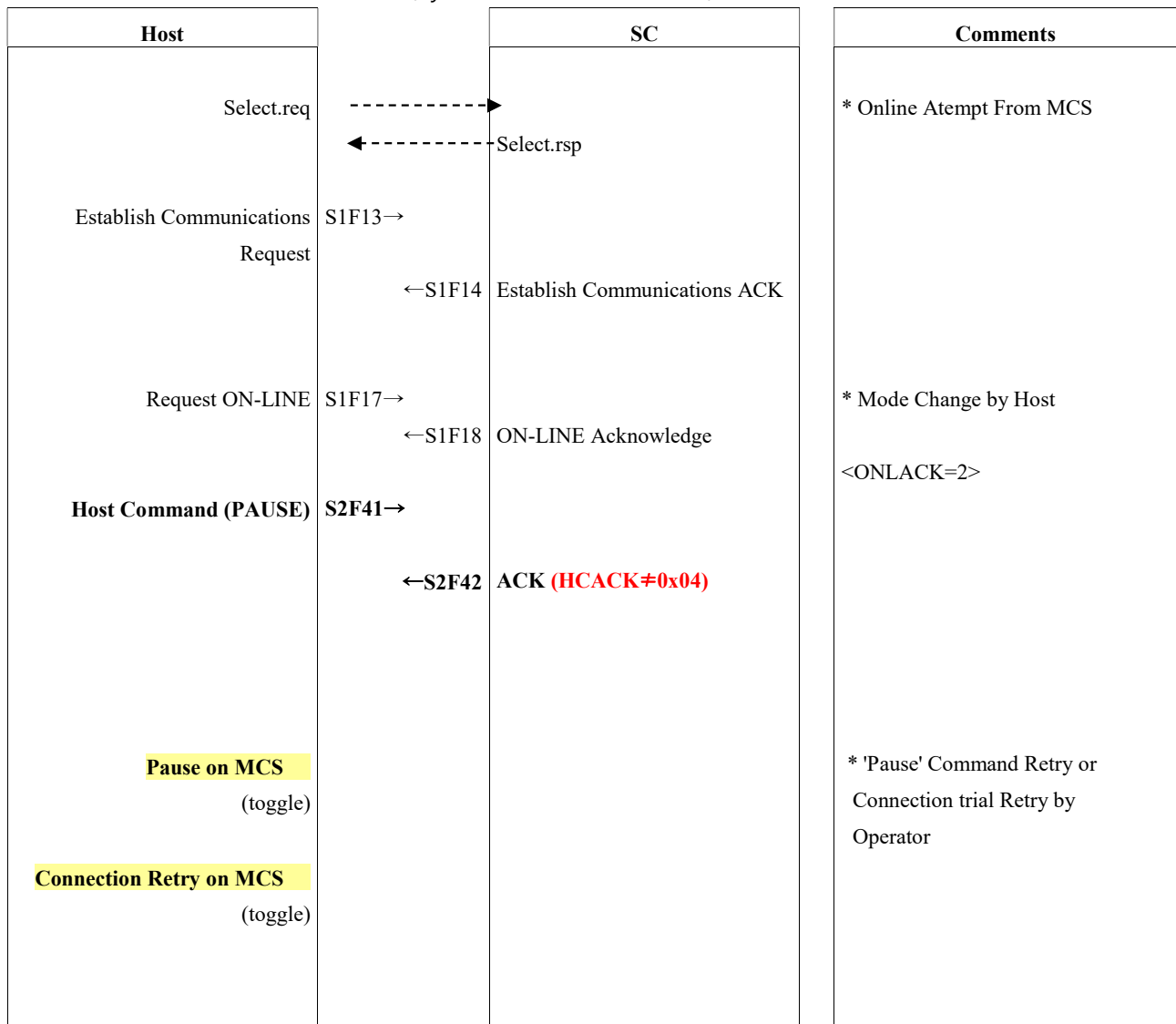
1.3. ONLINE → OFFLINE (By Host)

Host		TSC	Comments
Disconnect on MCS (toggle) Request OFF-LINE	S1F15→		* OFF-LINE Change by Host
	←S1F16	OFF-LINE Acknowledge	
	←S6F11	EquipmentOffline (CEID:1, REPORTID:1)	* OFF-LINE Change Report
ACK	S6F12→		

1.4. OFFLINE → ONLINE **Fail** (By Host, TSC State = Paused)



1.5. OFFLINE → ONLINE **Fail** (By Host, TSC State = Auto)



2. REMOTE COMMAND

2.1. ABORT Command

Host		TSC	Comments
Enhanced Remote Command	S2F49→		
	←S2F50	ACK (HCACK=0x04)	
	:		
Host Command (ABORT)	S2F41→		* Host want to abort
	←S2F42	ACK (HCACK=0x04)	command transferred to VHC
	←S6F11	TransferAbortInitiated (CEID:103, REPORTID:3)	
ACK	S6F12→		
	←S6F11	TransferAbortCompleted (CEID:101, REPORTID:3)	
ACK	S6F12→		

2.2. CANCEL Command

Host		TSC	Comments
Enhanced Remote Command	S2F49→		
	←S2F50	ACK (HCACK=0x04)	
	:		
Host Command (CANCEL)	S2F41→		* Host want to cancel command
	←S2F42	ACK (HCACK=0x04)	transferred to VHC
	←S6F11	TransferCancelInitiated (CEID:106, REPORTID:3)	
ACK	S6F12→		
	←S6F11	TransferCancelCompleted (CEID:104, REPORTID:3)	
ACK	S6F12→		

2.3. PAUSE Command

Host		TSC	Comments
Host Command (PAUSE)	S2F41→		
	←S2F42	ACK (HCACK=0x04)	* Host want to VHC state change to PAUSE
	←S6F11	TSCPauseInitiated (CEID:57, REPORTID:1)	* TSCState [Pausing]
ACK	S6F12→		
	←S6F11	TSCPauseCompleted (CEID:55, REPORTID:1)	* TSCState [Paused]
ACK	S6F12→		

2.4. RESUME Command

Host		TSC	Comments
Host Command (RESUME)	S2F41→		
	←S2F42	ACK (HCACK=0x04)	* Host want to VHC state change to AUTO
	←S6F11	TSCAutoCompleted (CEID:53, REPORTID:1)	* TSC State [Auto]
ACK	S6F12→		

2.5. UPDATE Command

Host		SC	Comments
Enhanced Remote Command	S2F49→		* Host want to update command transferred to SC
	←S2F50	ACK (HCACK=0x04)	
	:		
Host Command (UPDATE)	S2F41→		※ Update Command is accepted
	←S2F42	ACK (HCACK=0x04)	
	←S6F11	TransferUpdateCompleted (CEID:660, REPORTID:4)	* In the case of Updating data correctly
	ACK S6F12 →		
	←S6F11	DestUpdateCompleted (CEID:673, REPORTID:3)	* In the case of Updating the Dest
ACK	S6F12→		
	←S6F11	PriorityUpdateCompleted (CEID:674, REPORTID:3)	* In the case of Updating the Priority
ACK	S6F12→		
	←S6F11	TransferUpdateFailed (CEID:661, REPORTID:4)	* In the case that the updating command cannot be done even though it had been accepted.
ACK	S6F12→		
Host Command (UPDATE)	S2F41→		※ Update Command isn't accepted
	←S2F42	ACK (HCACK≠0x04)	

※ TransferUpdateCompleted message was separated into DestUpdateCompleted and PriorityUpdateCompleted. (for separating Dest and Priority). If both are changed at a time, DestUpdatedCompleted is used.

2.6. SCAN Command

Host	SC	Comments
Host Command (SCAN)	S2F41→	
	←S2F42 ACK (HCACK=0x04)	
	←S6F11 CarrierIDRead (CEID:251, REPORTID:11)	When SCAN order to carrier existing on port, and new carrier ID read.
ACK	S6F12→	ID Read Mismatch
	←S6F11 CarrierRemoved (CEID:152, REPORTID:4)	Old carrier ID on port (Scanning port location)
ACK	S6F12→	New carrier ID on port (Scanning port location)
	←S6F11	
ACK	S6F12→	
	←S6F11 CarrierIDRead (CEID:251, REPORTID:11)	When SCAN order to carrier existing on port, and carrier ID which currently exists on other port read.
ACK	S6F12→	ID Read Duplicate
	←S6F11 CarrierRemoved (CEID:152, REPORTID:4)	Old carrier ID on port (Scanning port location)
ACK	S6F12→	
	←S6F11 CarrierRemoved (CEID:152, REPORTID:4)	Old carrier ID on port (Port location which is same with currently read carrier ID)
ACK	S6F12→	
	←S6F11 CarrierInstalled (CEID:151, REPORTID:4)	New carrier ID on port (Scanning port location)
ACK	S6F12→	
	←S6F11 CarrierInstalled	UNKNOWN carrier ID on port

ACK	S6F12→	(CEID:151, REPORTID:4)	(Port location which is same with currently read carrier ID)
			When SCAN order to carrier existing on port, and same carrier ID (not duplicate, currently exists on scanning port) read.
	←S6F11	CarrierIDRead	ID Read Success
ACK	S6F12→	(CEID:251, REPORTID:11)	
			When SCAN order to empty port, and duplicate carrier ID to other port is read.
	←S6F11	CarrierIDRead	ID Read Duplicate
ACK	S6F12→	(CEID:251, REPORTID:11)	
	←S6F11	CarrierRemoved	From Existing position.
ACK	S6F12→	(CEID:152, REPORTID:4)	
	←S6F11	CarrierInstalled	Report with read CstID on new positon.
ACK	S6F12→	(CEID:151, REPORTID:4)	(Newly read CstID)
	←S6F11	CarrierInstalled	Report with Unknown CstID with existing position.
ACK	S6F12→	(CEID:151, REPORTID:4)	(Existing Carrier with duplicate CstID)

			When SCAN order to empty port, and new carrier ID is read.
	←S6F11	CarrierIDRead (CEID:251, REPORTID:11)	ID Read Success
ACK	S6F12→		
	←S6F11	CarrierInstalled (CEID:151, REPORTID:4)	New carrier ID on scanning port
ACK	S6F12→		
			Carrier exists on scanning port and scan fails, when RCP and MCS assume there is carrier on scanning port.
	←S6F11	CarrierIDRead (CEID:251, REPORTID:11)	ID Read Fail
ACK	S6F12→		
	←S6F11	CarrierRemoved (CEID:152, REPORTID:4)	Currently known carrier on scanning port.
ACK	S6F12→		
	←S6F11	CarrierInstalled (CEID:151, REPORTID:4)	UNKNOWN carrier.
ACK	S6F12→		
			Carrier doesn't exist on scanning port and scan fails, when RCP and MCS assume there is carrier on scanning port.
	←S6F11	CarrierIDRead (CEID:251, REPORTID:11)	ID Read Fail
ACK	S6F12→		
	←S6F11	CarrierRemoved (CEID:152, REPORTID:4)	Currently known carrier on scanning port.
ACK	S6F12→		

			Carrier exists on scanning port and scan fails, when RCP and MCS assume there is NO carrier on scanning port.	
		←S6F11	CarrierIDRead (CEID:251, REPORTID:11)	ID Read Fail
ACK	S6F12→			
		←S6F11	CarrierInstalled (CEID:151, REPORTID:4)	UNKNOWN carrier.
ACK	S6F12→			
		←S6F11	CarrierIDRead (CEID:251, REPORTID:11)	Carrier doesn't exist on scanning port and scan fails, when RCP and MCS assume there is NO carrier on scanning port.
ACK	S6F12→			ID Read Fail

2.7. Carrier Install Command (Used for Buffer Only)

Normal Case

Host		SC	Comments
Host Command (INSTALL)	S2F41→		
	←S2F42	ACK (HCACK=0x04)	
	←S6F11	CarrierInstalled (CEID:151, REPORTID:4)	
ACK	S6F12→		

Abnormal Case

Host	SC	Comments
Host Command (SCAN) S2F41→	←S2F42 ACK (HCACK=0x02)	Duplicate CstID or CarrierType is received. * If duplicate CstID occurs, RCP system rejects Install Command.

2.8. Carrier Remove Command (Used for Buffer Only)

Host	SC	Comments
Host Command (REMOVE) S2F41→	←S2F42 ACK (HCACK=0x04)	
	←S6F11 CarrierRemoved (CEID:152, REPORTID:4)	
ACK S6F12→		

3. VHC UNIT EVENT

3.1. VHC Installed

Host	TSC	Comments
	←S6F11 Vehicle Installed (CEID:208, REPORTID:8)	
ACK S6F12→		* VHC Controller report information involved vehicle used or unused

3.2. VHC Removed

Host	TSC	Comments
	←S6F11 Vehicle Removed (CEID:209, REPORTID:8)	
ACK S6F12→		* VHC Controller report information involved vehicle used or unused

3.3. PortInService

Host		TSC	Comments
	←S6F11	PortInService (CEID:261, REPORTID:10)	* VHC Controller report about port status in Controller
ACK	S6F12→		

3.4. PortOutOfService

Host		TSC	Comments
	←S6F11	PortOutOfService (CEID:260, REPORTID:10)	* VHC Controller report about port status in Controller
ACK	S6F12→		

3.4. VehicleCannotCallSet

Host		TSC	Comments
	←S6F11	VehicleCannotCallSet (CEID:211, REPORTID:8)	* VHC Controller report aboutVehicleCannotCallSetting
ACK	S6F12→		

3.5. VehicleCannotCallClear

Host		TSC	Comments
	←S6F11	VehicleCannotCallClear (CEID:212, REPORTID:8)	* VHC Controller report aboutVehicleCannotCall Clear
ACK	S6F12→		

3.5. Vehicle Blocking Started/Ended

Host		TSC	Comments
	←S6F11	VehicleBlockingStarted (CEID:671, REPORTID:32)	* VHC is blocked by another vehicle
ACK	S6F12→		
	:		
	←S6F11	VehicleBlockingEnded (CEID:672, REPORTID:32)	* Blocking is cleared
ACK	S6F12→		

3.6. Monitored Vehicle Event

Host		TSC	Comments
	←S6F11	MonitoredVehicles	* VHC reports MonitoredVehicles Event periodically to MCS.
ACK	S6F12→		<L[7] <UnitID> <VehicleCurrentDomain> <VehicleNextDomain> <AlarmID> <AlarmText> <VehicleCommunicationState> <MainteState> >

3.7. Vehicle Charge [AGV ONLY]

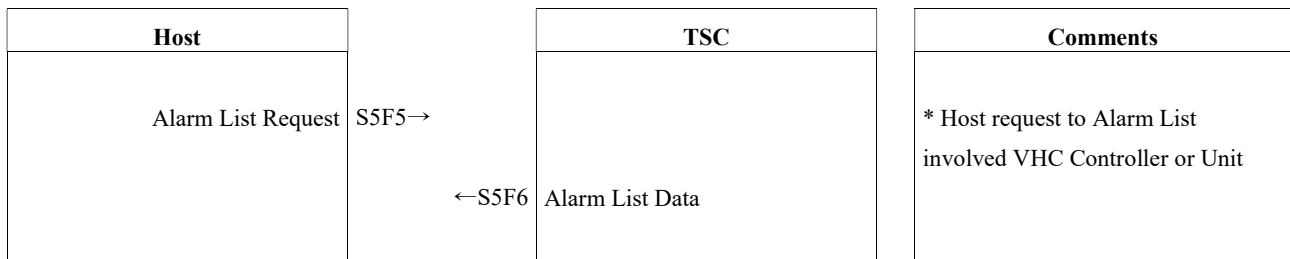
Host		TSC	Comments
TSC gives Charging Command for chargeable VHC			
		←S6F11 VehicleChargeSTStarted (CEID:662, REPORTID:31)	* VHC starts to move towards Charge Station
ACK	S6F12→		
		←S6F11 VehicleCharging (CEID:664, REPORTID:31)	* VHC arrives at Charge Station and starts Charging
ACK	S6F12→		
		←S6F11 VehicleNotCharged (CEID:665, REPORTID:31)	* VHC finishes Charging
ACK	S6F12→		

3.8. Vehicle Charge Fail [AGV ONLY]

Host		TSC	Comments
		←S6F11 VehicleChargeSTStarted (CEID:662, REPORTID:31)	* VHC starts to move towards Charge Station
	S6F12→		
ACK		←S6F11 VehicleCharging (CEID:664, REPORTID:31)	* VHC arrives at Charge Station and starts Charging
	S6F12→		
ACK		←S6F11 VehicleChargeFailed (CEID:663, REPORTID:31)	* VHC reports Charging Failure
	S6F12→		
ACK		←S6F11 VehicleNotCharged (CEID:665, REPORTID:31)	
	S6F12→		
ACK			

4. INFORMATION REQUEST

4.1. Alarm List REQUEST



5. TRANSFER COMMAND

- Transfer command and transfer scenario will be executed at the same time even two order commands are in progress.

5.1. STK/EQ → STK/EQ

Host		TSC	Comments
Enhanced Remote Command	S2F49→		* 'STAGE' Command
	←S2F50	ACK (HCACK=0x04)	
Enhanced Remote Command	S2F49→		* 'TRANSFER' Command
	←S2F50	ACK (HCACK=0x04)	
	←S6F11	TransferInitiated	
ACK	S6F12→	(CEID:108, REPORTID:3)	*If command with different carrier type received, replace to newly received carrier type.
	←S6F11	VehicleAssigned	
ACK	S6F12→	(CEID:204, REPORTID:7)	
	←S6F11	VehicleArrived (Source)	
ACK	S6F12→	(CEID:201, REPORTID:5)	
	←S6F11	Transferring	
	S6F12→	(CEID:111, REPORTID:3)	
	←S6F11	VehicleAcquireStarted	
ACK	S6F12→	(CEID:202, REPORTID:6)	
	←S6F11	VehicleCarrierInstalled	
ACK	S6F12→	(CEID:153, REPORTID:16)	
	←S6F11	VehicleAcquireCompleted	
ACK	S6F12→	(CEID:203, REPORTID:6)	
	←S6F11	VehicleDeparted	
ACK	S6F12→	(CEID:205, REPORTID:5)	
			* Carrier loaded on VHC

Host	TSC		Comments
	←S6F11	VehicleArrived (Dest)	
ACK	S6F12→	(CEID:201, REPORTID:5)	
	←S6F11	VehicleDepositStarted	
ACK	S6F12→	(CEID:206, REPORTID:6)	
	←S6F11	VehicleCarrierRemoved	
ACK	S6F12→	(CEID:154, REPORTID:16)	
	←S6F11	VehicleDepositCompleted	
ACK	S6F12→	(CEID:207, REPORTID:6)	
	←S6F11	VehicleUnassigned	
ACK	S6F12→	(CEID:210, REPORTID:7)	
	←S6F11	TransferCompleted	
ACK	S6F12→	(CEID:107, REPORTID:2)	

5.2. Buffer→ Buffer

Host		TSC	Comments
Enhanced Remote Command	S2F49→		* 'STAGE' Command
	←S2F50	ACK (HCACK=0x04)	
Enhanced Remote Command	S2F49→		* 'TRANSFER' Command
	←S2F50	ACK (HCACK=0x04)	
	←S6F11	TransferInitiated	
ACK	S6F12→	(CEID:108, REPORTID:3)	*If command with different carrier type received, replace to newly received carrier type.
	←S6F11	VehicleAssigned	
ACK	S6F12→	(CEID:204, REPORTID:7)	
	←S6F11	VehicleArrived (Source)	
ACK	S6F12→	(CEID:201, REPORTID:5)	
	←S6F11	Transferring	
	S6F12→	(CEID:111, REPORTID:3)	
	←S6F11	VehicleAcquireStarted	
ACK	S6F12→	(CEID:202, REPORTID:6)	
	←S6F11	CarrierRemoved	
ACK	S6F12→	(CEID:152, REPORTID:4)	* Carrier unloaded frombuffer
	←S6F11	VehicleCarrierInstalled	
ACK	S6F12→	(CEID:153, REPORTID:16)	* Carrier loaded on VHC
	←S6F11	VehicleAcquireCompleted	
ACK	S6F12→	(CEID:203, REPORTID:6)	
	←S6F11	VehicleDeparted	
ACK	S6F12→	(CEID:205, REPORTID:5)	

5.3. Command UPDATE

Host		TSC	Comments
Enhanced Remote Command	S2F49→		(Source ="EQ(a) Port", Destination="EQ(b) Port")
	←S2F50	ACK (HCACK=0x04)	
	:		
	←S6F11	VehicleCarrierInstalled	
	ACK S6F12→	(CEID:153, REPORTID:16)	
	←S6F11	VehicleAcquireCompleted	
	ACK S6F12→	(CEID:203, REPORTID:6)	
	←S6F11	VehicleDeparted	
	ACK S6F12→	(CEID:205, REPORTID:5)	
Host Command (UPDATE)	S2F41→		* the new command towards the new destination (Source ="EQ(a) Port", Destination="EQ(c) Port") * In the case of Updating the Dest * In the case of Updating the Priority * thesubsequential scenario is the same as Normal Transferring Scenario.
	←S2F42	ACK (HCACK=0x04)	
	←S6F11	DestUpdateCompleted	
	ACK S6F12→	(CEID:673, REPORTID:3)	
	←S6F11	PriorityUpdateCompleted	
	ACK S6F12→	(CEID:674, REPORTID:3)	
	:		
	←S6F11	VehicleArrived (Dest)	
	ACK S6F12→	(CEID:201, REPORTID:5)	

5.4. VHC Controller MANUAL Transfer (STK/EQ → STK/EQ)

Host		TSC	Comments
		←S6F11 OperationInitiatedAction (CEID:254, REPORTID:14)	* In the case that RCP gives manual transfer command * CommandType="EQ_TRANSFER"
ACK	S6F12→		
		←S6F11 TransferInitiated (CEID:108, REPORTID:3)	
ACK	S6F12→		
		←S6F11 VehicleAssigned (CEID:204, REPORTID:7)	
ACK	S6F12→		
		←S6F11 VehicleArrived (Source) (CEID:201, REPORTID:5)	
ACK	S6F12→		
		←S6F11 Transferring (CEID:111, REPORTID:3)	
ACK	S6F12→		
		←S6F11 VehicleAcquireStarted (CEID:202, REPORTID:6)	*thesubsequential scenario is the same as Normal Transferring Scenario.
ACK	S6F12→		
		←S6F11 CarrierRemoved (CEID:152, REPORTID:4)	
ACK	S6F12→		
		←S6F11 VehicleCarrierInstalled (CEID:153, REPORTID:16)	
ACK	S6F12→		
		←S6F11 VehicleAcquireCompleted (CEID:203, REPORTID:6)	
ACK	S6F12→		
		←S6F11 VehicleDeparted (CEID:205, REPORTID:5)	
ACK	S6F12→		
	:		

5.5. VHC Controller MANUAL Transfer (Buffer→ Buffer)

Host		TSC	Comments
	←S6F11	OperationInitiatedAction	* In the case that RCP gives manual transfer command * CommandType="EQ_TRANSFER"
ACK	S6F12→	(CEID:254, REPORTID:14)	
	←S6F11	TransferInitiated	
ACK	S6F12→	(CEID:108, REPORTID:3)	
	←S6F11	VehicleAssigned	
ACK	S6F12→	(CEID:204, REPORTID:7)	
	←S6F11	VehicleArrived (Source)	
ACK	S6F12→	(CEID:201, REPORTID:5)	
	←S6F11	Transferring	
ACK	S6F12→	(CEID:111, REPORTID:3)	
	←S6F11	VehicleAcquireStarted	*the subsequential scenario is the same as Normal Transferring Scenario.
ACK	S6F12→	(CEID:202, REPORTID:6)	
	←S6F11	VehicleCarrierInstalled	
ACK	S6F12→	(CEID:153, REPORTID:16)	
	←S6F11	VehicleAcquireCompleted	
ACK	S6F12→	(CEID:203, REPORTID:6)	
	←S6F11	VehicleDeparted	
ACK	S6F12→	(CEID:205, REPORTID:5)	
	:		

6. COMMAND DELETE

6.1. CANCEL (By Host)

Host		TSC	Comments
Enhanced Remote Command Host Command (CANCEL)	S2F49→		
		←S2F50 ACK (HCACK=0x04)	
		←S6F11 TransferInitiated (CEID:108, REPORTID:3)	
	ACK S6F12→		
		←S6F11 VehicleAssigned (CEID:204, REPORTID:7)	
	ACK S6F12→		
		←S2F42 ACK (HCACK=0x04)	
		←S6F11 TransferCancelInitiated (CEID:106, REPORTID:3)	
	ACK S6F12→		
		←S6F11 TransferCancelCompleted (CEID:104, REPORTID:3)	
	ACK S6F12→		
		←S6F11 VehicleUnassigned (CEID:210, REPORTID:7)	
	ACK S6F12→		

6.2. CANCEL (By TSC)

Host		TSC	Comments
Enhanced Remote Command	S2F49→		
		←S2F50 ACK (HCACK=0x04)	
		←S6F11 TransferInitiated (CEID:108, REPORTID:3)	
ACK	S6F12→		
		←S6F11 VehicleAssigned (CEID:204, REPORTID:7)	
ACK	S6F12→		
		Transfer Command Delete (toggle)	
		←S6F11 OperatorInitiatedAction (CEID:254, REPORTID:9)	* CommandType= "CANCEL"
ACK	S6F12→		
		←S6F11 TransferCancelInitiated (CEID:106, REPORTID:3)	
ACK	S6F12→		
		←S6F11 TransferCancelCompleted (CEID:104, REPORTID:3)	
ACK	S6F12→		
		←S6F11 VehicleUnassigned (CEID:210, REPORTID:7)	
ACK	S6F12→		

6.3 ABORT (Normal, By Host)

Host		TSC	Comments
Enhanced Remote Command	S2F49→		
		←S2F50 ACK (HCACK=0x04)	
		←S6F11 TransferInitiated (CEID:108, REPORTID:3)	
	ACK S6F12→ :	←S6F11 VehicleDeparted (CEID:205, REPORTID:5)	
	ACK S6F12→		
Host Command (ABORT)	S2F41→		
		←S2F42 ACK (HCACK=0x04)	
		←S6F11 TransferAbortInitiated (CEID:103, REPORTID:3)	
	ACK S6F12→		
		←S6F11 TransferAbortCompleted (CEID:101, REPORTID:3)	
	ACK S6F12→ :		
Enhanced Remote Command	S2F49→		
		←S2F50 ACK (HCACK=0x04)	

* VHC is in a standby status after
Abort Command is accepted

* a substitutive command is given
towards AbortLocation
(by MCS operator)

6.4. ABORT (Global Duplicate, By Host)

Host		TSC	Comments
Enhanced Remote Command	S2F49→		
		←S2F50 ACK (HCACK=0x04)	
		←S6F11 TransferInitiated (CEID:108, REPORTID:3)	
	ACK S6F12→ :	←S6F11 VehicleDeparted (CEID:205, REPORTID:5)	
	ACK S6F12→		
Host Command (ABORT)	S2F41→		
		←S2F42 ACK (HCACK=0x04)	
		←S6F11 TransferAbortInitiated (CEID:103, REPORTID:3)	
	ACK S6F12→		
		←S6F11 TransferAbortCompleted (CEID:101, REPORTID:3)	
	ACK S6F12→		
ABORT After CarrierInstalled On Vehicle			
		←S6F11 VehicleCarrierRemoved (CEID:154, REPORTID:16)	Original CarrierID Removed
	ACK S6F12→ :		
		←S6F11 VehicleCarrierInstalled (CEID:153, REPORTID:16)	Unknown CarrierID Installed
	ACK S6F12→		
Enhanced Remote Command	S2F49→		
		ACK (HCACK=0x04)	
		←S2F50	

* VHC is in a standby status after Abort Command is accepted

* a substitutive command is given towards Abort Location
(by MCS operator)

6.5. ABORT (FAIL, By Host)

Host	TSC	Comments
Enhanced Remote Command Host Command (ABORT)	S2F49→	
	←S2F50	ACK (HACK=0x04)
	:	
	←S6F11	VehicleArrived (Dest) (CEID:201, REPORTID:5)
	ACK S6F12→	
	←S6F11	VehicleDepositStarted (CEID:206, REPORTID:6)
	ACK S6F12→	
	S2F41→	
	←S2F42	ACK (HACK=0x04)
	←S6F11	TransferAbortInitiated (CEID:103, REPORTID:3)
ACK	S6F12→	
	←S6F11	TransferAbortFailed (CEID:102, REPORTID:3)
	S6F12→	
	←S6F11	VehicleCarrierRemoved (CEID:154, REPORTID:16)
	S6F12→	
	:	

* VHS continues the transfer after
Abort Failure

7. ALARM

7.1. Alarm Set/Clear (Regarding one unit alarm)

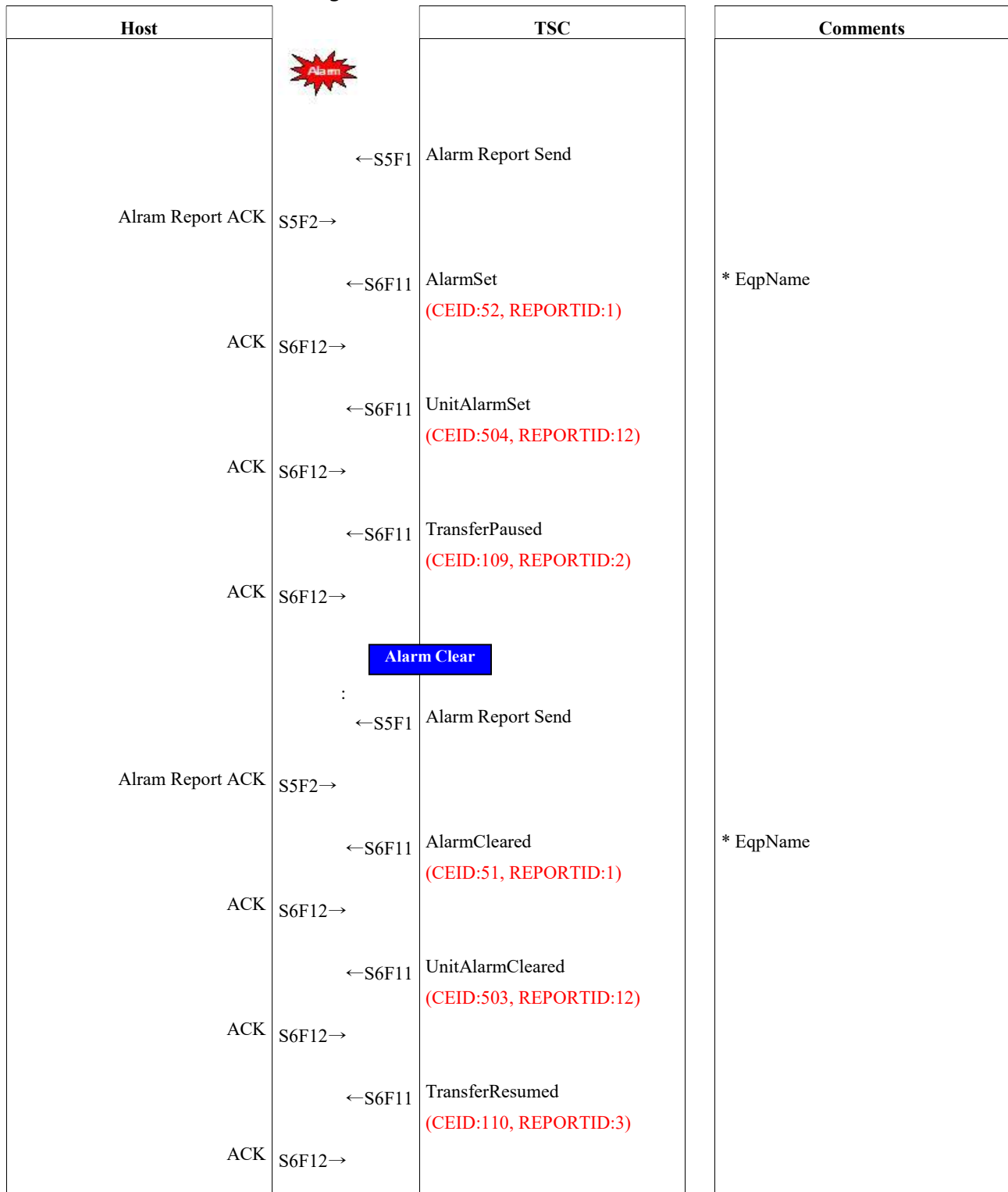
Host		TSC	Comments
			
		←S5F1 Alarm Report Send	
Alaram Report ACK	S5F2→		
		←S6F11 AlarmSet (CEID:52, REPORTID:1)	* EqpName
ACK	S6F12→		
		←S6F11 UnitAlarmSet (CEID:504, REPORTID:12)	
ACK	S6F12→		
		Alarm Clear	
	:		
		←S5F1 Alarm Report Send	
Alaram Report ACK	S5F2→		
		←S6F11 AlarmCleared (CEID:51, REPORTID:1)	* EqpName
ACK	S6F12→		
		←S6F11 UnitAlarmCleared (CEID:503, REPORTID:12)	
ACK	S6F12→		

7.2. Alarm Set/Clear (Regarding different alarms of more than two units)

Host		TSC	Comments
			
	←S5F1	Alarm Report Send	
Alarm Report ACK	S5F2→		
	←S6F11	AlarmSet	
ACK	S6F12→	(CEID:52, REPORTID:1)	
	←S6F11	UnitAlarmSet	
ACK	S6F12→	(CEID:504, REPORTID:12)	
	←S5F1	Alarm Report Send	
Alarm Report ACK	S5F2→		
	←S6F11	AlarmSet	
ACK	S6F12→	(CEID:52, REPORTID:1)	
	←S6F11	UnitAlarmSet	
ACK	S6F12→	(CEID:504, REPORTID:12)	
	Alarm Clear		
	←S5F1	Alarm Report Send	
Alarm Report ACK	S5F2→		
	←S6F11	AlarmCleared	
ACK	S6F12→	(CEID:51, REPORTID:1)	
	←S6F11	UnitAlarmCleared	
ACK	S6F12→	(CEID:503, REPORTID:12)	
	Alarm Clear		
	←S5F1	Alarm Report Send	
Alarm Report ACK	S5F2→		

Host			TSC	Comments
		←S6F11	AlarmCleared	
	ACK	S6F12→	(CEID:51, REPORTID:1)	
		←S6F11	UnitAlarmCleared	
	ACK	S6F12→	(CEID:503, REPORTID:12)	

7.3. Alarm Set/Clear Moving VHC



8. ALTERNATE TRANSFERRING

8.1. CST Loading Port Interlock Time Error (CST : PORT → VHC)

Host		TSC	Comments
Enhanced Remote Command	S2F49→		
		←S2F50 ACK (HACK=0x04)	
		←S6F11 TransferInitiated	
	ACK S6F12→	(CEID:108, REPORTID:3)	
		←S6F11 VehicleAssigned	
	ACK S6F12→	(CEID:204, REPORTID:7)	
		←S6F11 VehicleArrived (Source)	
	ACK S6F12→	(CEID:201, REPORTID:5)	
		←S6F11 Transferring	
	ACK S6F12→	(CEID:111, REPORTID:3)	
		←S6F11 VehicleAcquireStarted	
	S6F12→	(CEID:202, REPORTID:6)	
	ACK	Interlock Error	
	←S6F11 VehicleUnassigned		
ACK S6F12→	(CEID:210, REPORTID:7)		
	←S6F11 TransferCompleted		
ACK S6F12→	(CEID:107, REPORTID:2)		
	Once Retry		
Enhanced Remote Command	S2F49→		
		←S2F50 ACK (HACK=0x04)	
		←S6F11 TransferInitiated	
	ACK S6F12→	(CEID:108, REPORTID:3)	
	:		

* ResultCode =64

* CarrierLoc = "SourcePort"

* Source="SourcePort"

Destination="DestinationPort"

8.2. CST Unloading Port Interlock Time Error (CST : VHC → PORT)

Host		TSC	Comments
	:		*the previous scenario is the same as Normal Transferring Scenario.
	←S6F11	VehicleArrived (Dest)	
ACK	S6F12→	(CEID:201, REPORTID:5)	
	←S6F11	VehicleDepositStarted	
ACK	S6F12→	(CEID:206, REPORTID:6)	
	Interlock Error		
	←S6F11	VehicleUnassigned	
ACK	S6F12→	(CEID:210, REPORTID:7)	
	←S6F11	TransferCompleted	
ACK	S6F12→	(CEID:107, REPORTID:2)	
	Once Retry		* ResultCode =64 * CarrierLoc = "VHC UNIT NAME"
Enhanced Remote Command	S2F49→		
	←S2F50	ACK (HACK=0x04)	
	←S6F11	TransferInitiated	
ACK	S6F12→	(CEID:108, REPORTID:3)	* Source="VHC UNIT NAME" Destination="Previous Destination"
	←S6F11	VehicleAssigned	
ACK	S6F12→	(CEID:204, REPORTID:7)	
	←S6F11	Transferring	
ACK	S6F12→	(CEID:111, REPORTID:3)	

Host	TSC		Comments
	←S6F11	VehicleArrived (Dest)	
ACK	S6F12→	(CEID:201, REPORTID:5)	
	←S6F11	VehicleDepositStarted	
ACK	S6F12→	(CEID:206, REPORTID:6)	
	:		* a substitutive command is given towards AbortLocation after one repeated attempt (Source="VHC Unit Name" Destination="AbortLocation")

8.3. CST Unloading Port Interlock Time Error (In the case of All AbortLocation Failure)

Host		TSC	Comments
	:		
	←S6F11	VehicleArrived (Dest)	
ACK	S6F12→	(CEID:201, REPORTID:5)	* CST Unloading Port Interlock Time Error → moves to AbortLocation
	←S6F11	VehicleDepositStarted	
ACK	S6F12→	(CEID:206, REPORTID:6)	
		Interlock Error	
	←S6F11	VehicleUnassigned	
ACK	S6F12→	(CEID:210, REPORTID:7)	
	←S6F11	TransferCompleted	
ACK	S6F12→	(CEID:107, REPORTID:2)	
	:		
Enhanced Remote Command	S2F49→		* MCS gives a new command towards AbortLocation without retry (Source="VHC Unit Name" Destination="AbortLocation")
	←S2F50	ACK (HCACK=0x04)	
	:		
	←S6F11	VehicleArrived (Dest)	
ACK	S6F12→	(CEID:201, REPORTID:5)	
	←S6F11	VehicleDepositStarted	
ACK	S6F12→	(CEID:206, REPORTID:6)	
		Interlock Error	
	←S6F11	VehicleUnassigned	
ACK	S6F12→	(CEID:210, REPORTID:7)	
	←S6F11	TransferCompleted	
ACK	S6F12→	(CEID:107, REPORTID:2)	
No more AbortLocation, then MCS doesn't give any commandMCS			
	←S6F11	UnitAlarmSet	
ACK	S6F12→	(CEID:504, REPORTID:12)	doesn't have a command (Alarm Text : All Port Loading Fail - Manual Handling)

9. COMMAND

9.1. Duplication Command (Same Carrier)

Host		TSC	Comments
Enhanced Remote Command	S2F49→		
		←S2F50 ACK (HCACK=0x04)	
		←S6F11 TransferInitiated (CEID:108, REPORTID:3)	
	ACK S6F12→		
		←S6F11 VehicleAssigned (CEID:204, REPORTID:7)	
	ACK S6F12→		
		←S6F11 VehicleArrived (Source) (CEID:201, REPORTID:5)	
	ACK S6F12→		
		←S6F11 Transferring (CEID:111, REPORTID:3)	
	ACK S6F12→		
Enhanced Remote Command	S2F49→		
		←S2F50 ACK (HCACK=0x05)	
		←S6F11 VehicleAcquireStarted (CEID:202, REPORTID:6)	
	ACK S6F12→		
	:		* Either Cassette-ID orCommandID duplication check * 0x05 : rejected, already requested * The original command is carry out

9.2. No Command

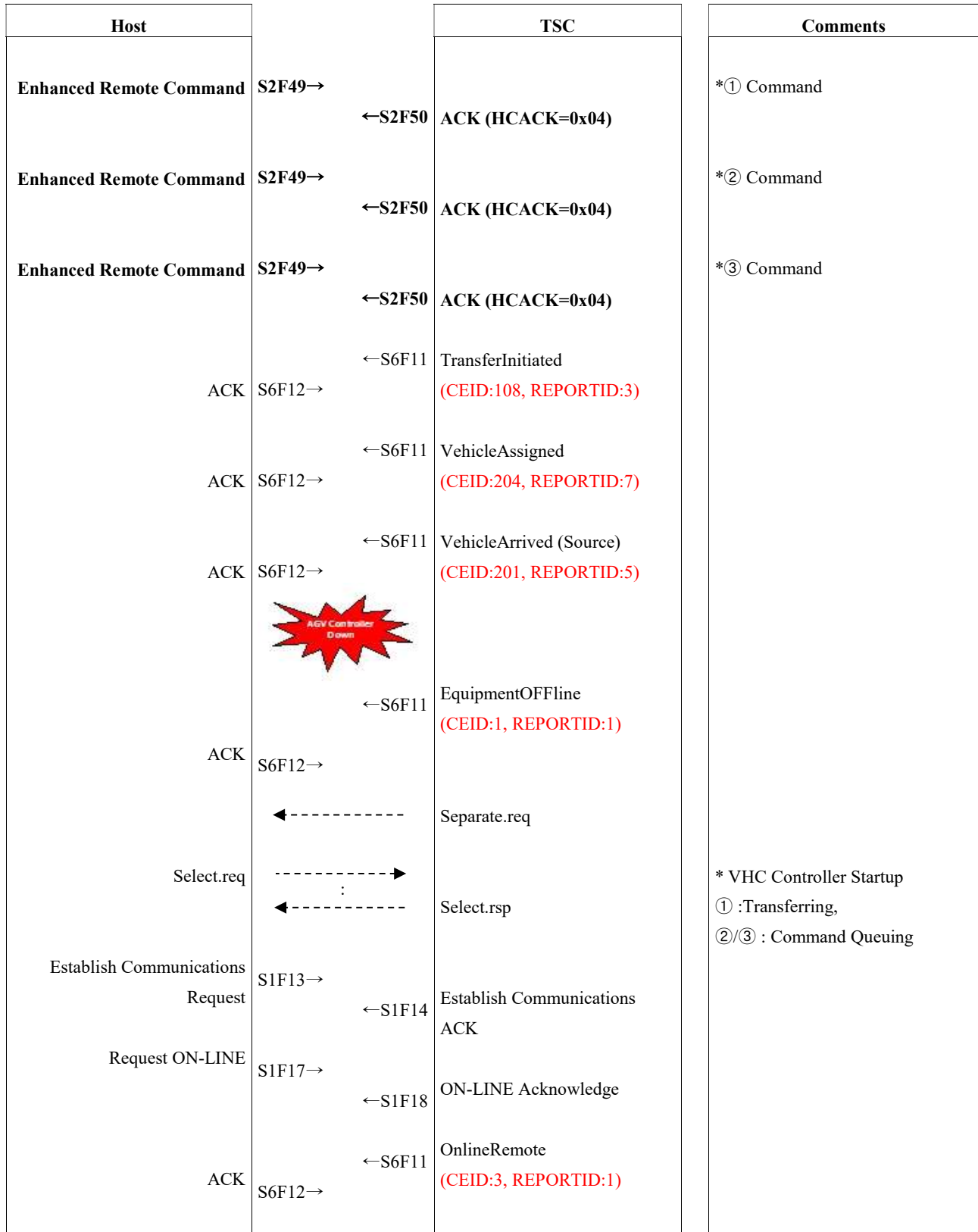
Host		TSC		Comments
In any case				
Enhanced Remote Command				*In the case that VHS is carrying a CST, but doesn't have a command (Alarm Text : All Port Loading Fail - Manual Handling)
	S2F49→ ←S2F50 :	ACK (HACK=0x04)		* MCS gives a new command or a manual order is given by TSC
ACK	S6F12→	Manual Transfer (on TSC) (toggle) ←S6F11 OperatorInitiatedAction (CEID:254, REPORTID:9) S6F12→		* CommandType= "EQ_TRNSFER"
ACK	S6F12→	←S6F11 TransferInitiated (CEID:108, REPORTID:3)		* thesubsequential scenario is the same as Normal Transferring Scenario.
ACK	S6F12→	←S6F11 VehicleAssigned (CEID:204, REPORTID:7)		
	:			

9.3. Duplication Command (Same Port)

Host		TSC	Comments
Enhanced Remote Command	S2F49→		
	←S2F50	ACK (HACK=0x04)	* Carrier : 1234 CommandId : 1234ABC Destination : A Port
	←S6F11	TransferInitiated (CEID:108, REPORTID:3)	
ACK	S6F12→		
RCP is carrying 1234 CST or in a stanby status.			
Enhanced Remote Command	S2F49→		* Carrier : 5678 CommandId : 5678DEF Destination: A Port
	←S2F50	ACK (HACK=0x08)	* 0x08 : Port reserved by other Carrier Transferring (HACK=08 occurs when it comes to A Port)
	:		
	←S6F11	Transferring (CEID:111, REPORTID:3)	* The transfer of already-existing- ID's carrier is carry out
ACK	S6F12→		

There is a case that OHS / OHT carries CSTs to the same port
The scenario above is deleted (Since 2014.10.30)

10. VHC CONTROLLER DOWN → UP



Host		TSC	Comments
	←S6F11	TSCAutoInitiated	
ACK	S6F12→	(CEID:54, REPORTID:1)	
	←S6F11	TSCPaused	
ACK	S6F12→	(CEID:56, REPORTID:1)	
Data and Time Set Request	S2F31→		
	←S2F32	Data and Time Set Request ACK	
Selected Equipment Status	S1F3→		
Request	←S1F4	Selected Equipment Status Data	※ Controller sends①,②,③
	←S6F11	Transferring	
ACK	S6F12→	(CEID:111, REPORTID:3)	* Controller continues operating
	:		
Host Command (RESUME)	S2F41→		
	←S2F42	ACK (HCACK=0x04 / 0x07)	
	←S6F11	TSCAutoCompleted	
ACK	S6F12→	(CEID:53, REPORTID:1)	
	:		
	←S6F11	VehicleAcquireStarted	
ACK	S6F12→	(CEID:202, REPORTID:6)	

11. Enable/Disable Event Report

11.1. Enable/Disable Event Request

Host		TSC	Comments
Enable Disable Event Request	S2F37→		
	←S2F38	ACK (ERACK=0x00)	<L [2] <BOOLEAN F > <L [1] <U2 671 > > → No Report for CEID 671
Enable Disable Event Request	S2F37→		
	←S2F38	ACK (ERACK=0x00)	<L [2] <BOOLEAN T > <L [1] <U2 671 > > → Report for CEID 671
	:		

※ CEID Enable,Disable requests are handled one by one (since : 2014.06.01)

12. DATE AND TIME SET

12.1. Date and Time Set

Host		TSC	Comments
Date and Time Set Data	S2F31→		* TSC sets date and time
	←S2F32	Date and Time Set ACK	

[How to synchronizetime between Host and TSC]

- 1. To make the menu in MCS UI for synchronizing time
- 2. To scheduler that MCS gives synchronizing msgat every 8 O'clock

13. BCR NG [Vehicle(include RFID Module) Only]

13.1. BCR Result (Sucess)

Host		TSC	Comments
	:		* the previous scenario is the same as Normal Transferring Scenario.
	←S6F11	VehicleCarrierInstalled (CEID:153, REPORTID:16)	
ACK	S6F12→		
	←S6F11	VehicleAcquireCompleted (CEID:203, REPORTID:6)	
ACK	S6F12→		* No report BCR Success
	:		* thesubsequential scenario is the same as Normal Transferring Scenario.

13.2. BCR Result (Failure)

Host		TSC	Comments
	:		* the previous scenario is the same as Normal Transferring Scenario.
	←S6F11	VehicleCarrierInstalled (CEID:153, REPORTID:16)	
ACK	S6F12→		
	←S6F11	VehicleAcquireCompleted (CEID:203, REPORTID:6)	
ACK	S6F12→		
	←S6F11	CarrierIDRead (CEID:251, REPORTID:11)	* IDReadStatus=1 : Failure
ACK	S6F12→		
	←S6F11	TransferCompleted (CEID:107, REPORTID:2)	* ResultCode = 6
ACK	S6F12→		
	←S6F11	VehicleCarrierRemoved (CEID:154, REPORTID:16)	* Original CarrierID Removal
ACK	S6F12→		
	←S6F11	VehicleCarrierInstalled (CEID:153, REPORTID:16)	* UNKNOWNID Creation (CarrierID : UNKNOWN+EQPNAME +YYMMDDhmmss+1Sec/100000)
ACK	S6F12→		
	:		* VHC waits for a new command
UNK Carrier			
Enhanced Remote Command	S2F49→		* MCS transfer UNKNOWN Carrier to AbortLocation
	←S2F50	ACK (HCACK=0x04)	

13.3. BCR Result (Duplicate)

Host		TSC	Comments
	:		* the previous scenario is the same as Normal Transferring Scenario.
	←S6F11	VehicleCarrierInstalled (CEID:153, REPORTID:16)	
ACK	S6F12→		
	←S6F11	VehicleAcquireCompleted (CEID:203, REPORTID:6)	
ACK	S6F12→		
	←S6F11	CarrierIDRead (CEID:251, REPORTID:11)	* IDReadStatus=2 : Duplicate
ACK	S6F12→		
	←S6F11	TransferCompleted (CEID:107, REPORTID:2)	* ResultCode =4
ACK	S6F12→		
	←S6F11	VehicleCarrierRemoved (CEID:154, REPORTID:16)	* Original CarrierID Removal
ACK	S6F12→		
	←S6F11	VehicleCarrierInstalled (CEID:153, REPORTID:16)	* UNKNOWNID Creation (CarrierID : UNKNOWN+EQPNAME +YYMMDDhhmmss+1Sec/100000)
ACK	S6F12→		
	:		* VHC waits for a new command
UNK Carrier			
Enhanced Remote Command	S2F49→		* MCS transfer UNKNOWN Carrier to AbortLocation
	←S2F50	ACK (HCACK=0x04)	

13.4. BCR Result (Mismatch)

Host		TSC	Comments
	:		*the previous scenario is the same as Normal Transferring Scenario.
	←S6F11	VehicleCarrierInstalled (CEID:153, REPORTID:16)	
ACK	S6F12→		
	←S6F11	VehicleAcquireCompleted (CEID:203, REPORTID:6)	
ACK	S6F12→		
	←S6F11	CarrierIDRead (CEID:251, REPORTID:11)	*IDReadStatus=3 : Mismatch
ACK	S6F12→		
	←S6F11	TransferCompleted (CEID:107, REPORTID:2)	*ResultCode =5
ACK	S6F12→		
	←S6F11	VehicleCarrierRemoved (CEID:154, REPORTID:16)	*Original CarrierID Removal
ACK	S6F12→		
	←S6F11	VehicleCarrierInstalled (CEID:153, REPORTID:16)	* Read ID is created
ACK	S6F12→		
	:		* VHC waits for a new command
New Carrier			
Enhanced Remote Command	S2F49→		* MCS transfer UNKNOWN Carrier to AbortLocation
	←S2F50	ACK (HCACK=0x04)	

14. CARRIER REMOVED

14.1. Carrier on VHC Removed with MU

Host		TSC	Comments
		AGV Alarm	
		CST removal by MU	
		Transfer Command Delete (toggle)	
	←S6F11	OperatorInitiatedAction (CEID:254, REPORTID:9)	*CommandType= "ABORT"
ACK	S6F12→		
	←S6F11	TransferAbortInitiated (CEID:103, REPORTID:3)	*CarrierLoc = "VHC Unit Name"
ACK	S6F12→		
	←S6F11	TransferAbortCompleted (CEID:101, REPORTID:3)	
ACK	S6F12→		
		CarrierRemoved (on MCP) (toggle)	
	←S6F11	VehicleCarrierRemoved (CEID:154, REPORTID:16)	
ACK	S6F12→		

14.2. Carrier Removed after Vehicle Removed

Host		TSC	Comments
		For AGV maintenance	
	←S6F11	Vehicle Removed (CEID:209, REPORTID:8)	
ACK	S6F12→		*VHS and Command which is related to VHS are removed at the same time
	←S6F11	Transfer Command Delete OperatorInitiatedAction (CEID:254, REPORTID:9)	*CommandType= "ABORT"
ACK	S6F12→		
	←S6F11	TransferAbortInitiated (CEID:103, REPORTID:3)	
ACK	S6F12→		
	←S6F11	TransferAbortCompleted (CEID:101, REPORTID:3)	*CarrierLoc = "VHCUnitName"
ACK	S6F12→		
		CarrierRemoved (on MCP) (toggle)	* The manual removal by OCS operator or The automatic removal are necessary
	←S6F11	VehicleCarrierRemoved (CEID:154, REPORTID:16)	
ACK	S6F12→		* The information of CST in VHC,MCS,MCP has to be removed
	:		
	←S6F11	VehicleInstalled (CEID:208, REPORTID:8)	
ACK	S6F12→		

14.3. Vehicle with Carrier Install after Vehicle Removed

Host		TSC	Comments
		For AGV maintenance	
		←S6F11 Vehicle Removed	
ACK	S6F12→	(CEID:209, REPORTID:8)	*VHS and Command which is related to VHS are removed at the same time
			
		Transfer Command Delete	
		←S6F11 OperatorInitiatedAction	
ACK	S6F12→	(CEID:254, REPORTID:9)	*CommandType= "ABORT"
		←S6F11 TransferAbortInitiated	
ACK	S6F12→	(CEID:103, REPORTID:3)	
	:	←S6F11 TransferAbortCompleted	
ACK	S6F12→	(CEID:101, REPORTID:3)	*CarrierLoc = "VHC Unit Name"
	:	←S6F11 VehicleInstalled	
ACK	S6F12→	(CEID:208, REPORTID:8)	
			*In the case that VHS is carrying a CST, but doesn't have a command
		Manual Transfer (on MCP)	(Alarm Text : All Port Loading Fail - Manual Handling)
		(toggle)	
		←S6F11 OperatorInitiatedAction	
ACK	S6F12→	(CEID:254, REPORTID:9)	*CommandType= "EQ_TRNSFER"
		←S6F11 TransferInitiated	
ACK	S6F12→	(CEID:108, REPORTID:3)	
		←S6F11 VehicleAssigned	
ACK	S6F12→	(CEID:204, REPORTID:7)	*thesubsequential scenario is the same as Normal Transferring Scenario.
	:		

15. ABNOMAL SITUATION

15.1. Vehicle Empty Retrieval

Host	TSC	Comments
Enhanced Remote Command	S2F49→	
	←S2F50 ACK (HCACK=0x04)	
	←S6F11 TransferInitiated	
ACK	S6F12→ (CEID:108, REPORTID:3)	
	←S6F11 VehicleAssigned	
ACK	S6F12→ (CEID:204, REPORTID:7)	
	←S6F11 VehicleArrived (Source)	
ACK	S6F12→ (CEID:201, REPORTID:5)	
	←S6F11 Transferring	
ACK	S6F12→ (CEID:111, REPORTID:3)	
	←S6F11 VehicleAcquireStarted	
	S6F12→ (CEID:202, REPORTID:6)	
ACK	Empty Retrieval Error	
	←S6F11 TransferPaused	
	(CEID:109, REPORTID:2)	
ACK	S6F12→	
ABORT	S2F41→	
	←S2F42 ACK	
	←S6F11 Carrier Removed	
ACK	S6F12→ (CEID:152, REPORTID:4)	
	←S6F11 Carrier Installed	
ACK	S6F12→ (CEID:151, REPORTID:4)	- Unknown Carrier Installed report from UTB
	←S6F11 Transfer Abort Initiated	
ACK	S6F12→ (CEID:103, REPORTID:3)	
	←S6F11 Transfer Abort Completed	
ACK	S6F12→ (CEID:101, REPORTID:3)	
	←S6F11 VehicleUnassigned	
	S6F12→ (CEID:210, REPORTID:7)	

15.2. Vehicle Double Storage

Host		TSC	Comments
	←S6F11	VehicleArrived (Dest)	
ACK	S6F12→	(CEID:201, REPORTID:5)	
	←S6F11	VehicleDepositStarted	
ACK	S6F12→	(CEID:206, REPORTID:6)	
	DoubleStore		
	Error		
	←S6F11	TransferPaused	
ACK	S6F12→	(CEID:109, REPORTID:2)	
ABORT	S2F41→		
	←S2F42	ACK	
	←S6F11	Carrier Installed	
ACK	S6F12→	(CEID:151, REPORTID:4)	
	←S6F11	Transfer Abort Initiated	
ACK	S6F12→	(CEID:103, REPORTID:3)	
	←S6F11	Transfer Abort Completed	
ACK	S6F12→	(CEID:101, REPORTID:3)	
	New Order		
		ACK (HCACK=0x04)	
Enhanced Remote Command	S2F49→		
	←S2F50	TransferInitiated	
		(CEID:108, REPORTID:3)	
	←S6F11		
ACK	S6F12→		
	:		

15.3. STB Double Storage

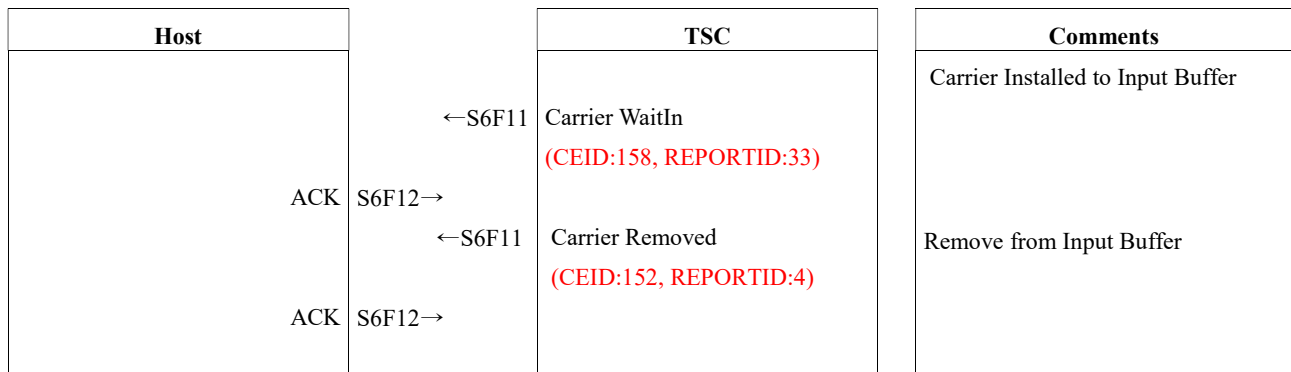
Host		TSC	Comments
	←S6F11	VehicleArrived (Dest)	
ACK	S6F12→	(CEID:201, REPORTID:5)	
	←S6F11	VehicleDepositStarted	
ACK	S6F12→	(CEID:206, REPORTID:6)	Vehicle 1 arrived
	DoubleStore		
	Error		
	←S6F11	TransferPaused	
ACK	S6F12→	(CEID:109, REPORTID:2)	
ABORT	S2F41→		
	←S2F42	ACK	
	←S6F11	Carrier Installed	
ACK	S6F12→	(CEID:151, REPORTID:4)	Unknown Carrier Installed report from STB
	←S6F11	Transfer Abort Initiated	
ACK	S6F12→	(CEID:103, REPORTID:3)	
	←S6F11	Transfer Abort Completed	
ACK	S6F12→	(CEID:101, REPORTID:3)	
	New Order		
Enhanced Remote Command	S2F49→		New Order Create From Vehicle 1 to another STB
	←S2F50	ACK (HCACK=0x04)	
	←S6F11	TransferInitiated	
ACK	S6F12→	(CEID:108, REPORTID:3)	Vehicle 1 receives new order (To Order)
	New Order		
	:		
Enhanced Remote Command	S2F49→		New Order Create for Unknown Carrier to STB which can read Carrier ID.
	←S2F50	ACK (HCACK=0x04)	
	←S6F11	TransferInitiated	
ACK	S6F12→	(CEID:108, REPORTID:3)	Another Vehicle X receives new order

15.4. STB Double Storage (Manual Order Create)

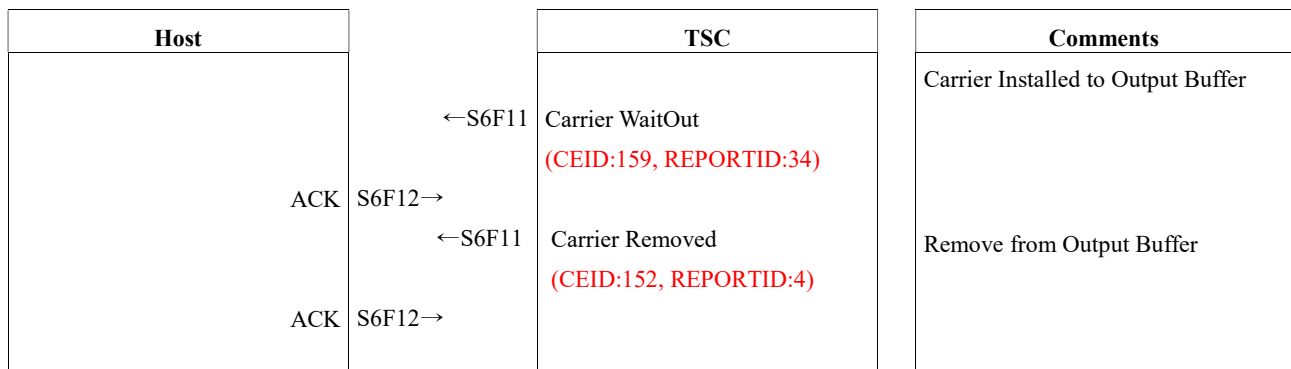
Host		TSC	Comments
	←S6F11	Carrier Installed	Unknown Carrier Installed from STB
ACK	S6F12→	(CEID:151, REPORTID:4)	which has not Carrier ID Reader
			*If Host couldn't create new order for unknown carrier, operator need to make new order manually
	←S6F11	OperationInitiatedAction	Operator Create New Order for
ACK	S6F12→	(CEID:254, REPORTID:14)	Unknown Carrier to STB which has
			Carrier ID Reader
	←S6F11	TransferInitiated	
ACK	S6F12→	(CEID:108, REPORTID:3)	
	...	...	
	←S6F11	TransferCompleted	Transfer Complete
ACK	S6F12→	(CEID:107, REPORTID:2)	
	←S6F11	Carrier Installed	Carrier Installed Report with Read
ACK	S6F12→	(CEID:151, REPORTID:4)	Carrier ID from STB.
	New Order		
	:		
Enhanced Remote Command	S2F49→		New Order Create for Read Carrier
	←S2F50	ACK (HCACK=0x04)	
	←S6F11	TransferInitiated	
ACK	S6F12→	(CEID:108, REPORTID:3)	

16. Wait In / Wait Out Event (Manual Port)

16.1. Wait In Event



16.2. Wait Out Event



17. BCR Read (Reading Station Only)

17.1. BCR Result (Sucess)

Host		TSC	Comments
	:		
	←S6F11	TransferInitiated (CEID:108, REPORTID:3)	* the previous scenario is the same as Normal Transferring Scenario.
ACK	S6F12→		
	←S6F11	VehicleArrived (Dest) (CEID:201, REPORTID:5)	
ACK	S6F12→		
	←S6F11	VehicleDepositStarted (CEID:206, REPORTID:6)	
ACK	S6F12→		
	←S6F11	VehicleCarrierRemoved (CEID:154, REPORTID:16)	
ACK	S6F12→		* Order of Vehicle DepositCompleted, Carrier Installed, CarrierIDRead Message can be changed
	←S6F11	VehicleDepositeCompleted (CEID:207, REPORTID:6)	
ACK	S6F12→		
	←S6F11	Carrier Installed (CEID:151, REPORTID:4)	* Read Success
ACK	S6F12→		
	←S6F11	CarrierIDRead (CEID:251, REPORTID:11)	* Install Read Cst ID
ACK	S6F12→		
	←S6F11	VehicleUnassigned (CEID:210, REPORTID:7)	* IDReadStatus 0 : Success 1 : Fail 2 : Duplicate 3 : Mismatch
ACK	S6F12→		
	←S6F11	TransferCompleted (CEID:107, REPORTID:2)	
ACK	S6F12→		

17.2. BCR Result (Failure)

Host		TSC	Comments
	:		
	←S6F11	TransferInitiated (CEID:108, REPORTID:3)	* the previous scenario is the same as Normal Transferring Scenario.
ACK	S6F12→		
	←S6F11	VehicleArrived (Dest) (CEID:201, REPORTID:5)	* IDReadStatus 0 : Success 1 : Fail 2 : Duplicate 3 : Mismatch
ACK	S6F12→		
	←S6F11	VehicleDepositStarted (CEID:206, REPORTID:6)	
ACK	S6F12→		
	←S6F11	VehicleCarrierRemoved (CEID:154, REPORTID:16)	* Order of Vehicle DepositCompleted, Carrier Installed, CarrierIDRead Message can be changed.
ACK	S6F12→		
	←S6F11	VehicleDepositeCompleted (CEID:207, REPORTID:6)	
ACK	S6F12→		
	←S6F11	Carrier Installed (CEID:151, REPORTID:4)	* Failure
ACK	S6F12→		
	←S6F11	CarrierIDRead (CEID:251, REPORTID:11)	* Install Unknown Cst ID
ACK	S6F12→		
	←S6F11	VehicleUnassigned (CEID:210, REPORTID:7)	
ACK	S6F12→		
	←S6F11	TransferCompleted (CEID:107, REPORTID:2)	* Result Code = 6
ACK	S6F12→		

17.3. BCR Result (Duplicate)

Host		TSC	Comments
	:		
	←S6F11	TransferInitiated (CEID:108, REPORTID:3)	* the previous scenario is the same as Normal Transferring Scenario.
ACK	S6F12→		
	←S6F11	VehicleArrived (Dest) (CEID:201, REPORTID:5)	* IDReadStatus 0 : Success 1 : Fail 2 : Duplicate 3 : Mismatch
ACK	S6F12→		
	←S6F11	VehicleDepositStarted (CEID:206, REPORTID:6)	
ACK	S6F12→		
	←S6F11	VehicleCarrierRemoved (CEID:154, REPORTID:16)	* Order of Vehicle DepositCompleted, Carrier Installed, CarrierIDRead Message can be changed.
ACK	S6F12→		
	←S6F11	VehicleDepositeCompleted (CEID:207, REPORTID:6)	
ACK	S6F12→		
	←S6F11	Carrier Removed (CEID:152, REPORTID:4)	* Existing duplicate CST ID from existing station.
ACK	S6F12→		
	←S6F11	Carrier Installed (CEID:151, REPORTID:4)	* Unknown CST ID
ACK	S6F12→		
	←S6F11	Carrier Installed (CEID:151, REPORTID:4)	* Install Read Cst ID from Reading Station
ACK	S6F12→		
	←S6F11	CarrierIDRead (CEID:251, REPORTID:11)	* Duplicate CST ID exists on RCP System.
ACK	S6F12→		
	←S6F11	VehicleUnassigned (CEID:210, REPORTID:7)	
ACK	S6F12→		
	←S6F11	TransferCompleted (CEID:107, REPORTID:2)	* Result Code = 4
ACK	S6F12→		

17.4. BCR Result (Mismatch)

Host		TSC	Comments
	:		
	←S6F11	TransferInitiated (CEID:108, REPORTID:3)	* the previous scenario is the same as Normal Transferring Scenario.
ACK	S6F12→		
	←S6F11	VehicleArrived (Dest) (CEID:201, REPORTID:5)	* IDReadStatus 0 : Success 1 : Fail 2 : Duplicate 3 : Mismatch
ACK	S6F12→		
	←S6F11	VehicleDepositStarted (CEID:206, REPORTID:6)	
ACK	S6F12→		
	←S6F11	VehicleCarrierRemoved (CEID:154, REPORTID:16)	* Order of Vehicle DepositCompleted, Carrier Installed, CarrierIDRead Message can be changed.
ACK	S6F12→		
	←S6F11	VehicleDepositeCompleted (CEID:207, REPORTID:6)	
ACK	S6F12→		
	←S6F11	Carrier Installed (CEID:151, REPORTID:4)	* Mismatch with order received from MCS
ACK	S6F12→		
	←S6F11	CarrierIDRead (CEID:251, REPORTID:11)	* Install Read Cst ID from Reading Station
ACK	S6F12→		
	←S6F11	VehicleUnassigned (CEID:210, REPORTID:7)	
ACK	S6F12→		
	←S6F11	TransferCompleted (CEID:107, REPORTID:2)	* Result Code = 5
ACK	:		